

KEERTHIRAJAN SENTHILKUMAR

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EDUCATION

The George Washington University

Master's in Computer Science (GPA: 3.84/4.0)

Aug 2024 - May 2026

Washington, D.C

- **Coursework:** Distributed Systems, Machine Learning, Neural Networks, Cloud Computing, Software Paradigms, Data Mining

Sri Venkateswara College of Engineering

Bachelor of Engineering, Computer Science (GPA: 3.54/4.0)

Aug 2019 - May 2023

Chennai, India

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, Kotlin, SQL
- **Cloud Platforms:** Azure ML, Azure OpenAI, AWS (EC2, S3, Lambda), Google Vertex AI, Firebase, BigQuery
- **AI/ML:** Foundational models, LoRA/PEFT, prompt engineering, LLM-as-judge evaluation frameworks, agentic AI patterns, LangChain, PyTorch, TensorFlow, semantic search, vector embeddings, retrieval ranking, transformer architectures
- **MLOps:** Model evaluation frameworks, ML monitoring, model versioning, offline evaluation pipelines, synthetic data generation, supervised fine-tuning (SFT), inference optimization, observability & telemetry
- **SW Engineering:** FastAPI, Docker, Kubernetes basics, Git, CI/CD pipelines, REST APIs, API design & security (OAuth 2.0 concepts), Infrastructure-as-Code, microservices architecture, system design

PROFESSIONAL EXPERIENCE

Apple | AIML Machine Learning Intern - Siri Core Modeling

Seattle, WA | May 2025 - Aug 2025

- Built large-scale evaluation framework for LLM-based tool selection ranking across 8 domains, identifying 60%+ failure rate in ranking quality on ambiguous queries, directly informing ranking model improvements
- Designed and implemented LLM-as-judge evaluation framework with embedding similarity metrics, reducing manual review cycles by 60% and enabling continuous automated model validation at scale
- Engineered vector embedding pipeline to filter 8,000+ BFCL tool candidates → 1,000 high-quality tool pairs for agentic LLM benchmarking and responsible deployment
- Quantified model robustness gaps in production systems: Gorilla v2 accuracy degradation from 83.5% → 33.3% on adversarial datasets, informing next-generation agentic architecture
- Designed evaluation metrics & testing strategies for multi-domain agentic systems with operational thinking for AI deployments

TCS iON | Machine Learning Intern

Chennai, India | Nov 2021 - Dec 2021

- Deployed production NLP pipeline for Twitter sentiment classification using Python, Flask, Docker, delivering real-time relevance scoring with 95%+ uptime, demonstrating search-quality thinking at scale
- Built RESTful APIs for text ingestion and ranking, handling 10K+ requests/day with automated monitoring and relevance metrics
- Improved classification accuracy by 18% through feature engineering (TF-IDF, n-grams) and continuous model updates on streaming data, similar to search ranking quality improvements
- Reduced inference latency by 40% by optimization and implemented pytest validation for CI, critical for low-latency search systems

AI / MACHINE LEARNING PROJECTS

BarterBrAI: Multi-Modal AI Barter Platform | Best Hack for Social Good - 1st place out of 80+ teams, 450+ participants

- Developed multi-modal agentic system using Gemini 2.5 Pro for intelligent item valuation, automated fairness assessment, and decision reasoning with confidence thresholds
- Implemented LLM-judged valuation logic within agent framework, incorporating safety guardrails and constrained decision boundaries to prevent fraudulent outcomes
- Architected Firebase Cloud Functions + Node.js microservices backend with event-driven workflows

Historical Event Narrator: Fine-Tuned LLM for Creative Narrative Generation

- Fine-tuned Mistral-7B with PyTorch using LoRA/PEFT supervised fine-tuning (SFT), achieving 100%+ ROUGE-L improvement and measurable quality gains (3/10 → 8/10 narrative quality)
- Engineered synthetic training dataset pipeline by transforming 1,200+ Wikipedia articles into curated narrative-twist pairs, demonstrating data-driven ML approach without large-scale labeled datasets
- Conducted A/B testing to measure model quality improvements, validating through LLM-as-judge evaluation across relevance, creativity, and quality metrics, demonstrating data-driven product development thinking
- Implemented guided inference with prompt injection and stopping criteria to reduce hallucinations and mode collapse
- Deployed Streamlit application with real-time model scoring and monitoring, enabling continuous performance assessment

Semantic Job-to-Problem Recommender System

- Built large-scale semantic retrieval system using Sentence Transformers with 384-d embeddings to match user queries (job descriptions) to 1,571 content items (LeetCode problems), achieving 0.65-0.75 relevance scoring via vector similarity
- Designed ranking pipeline that combined semantic matching with LLM-based re-ranking for improved relevance and personalization
- Deployed production retrieval system on Azure ML with FastAPI backend, demonstrating end-to-end search pipeline thinking from indexing to ranking to serving
- Integrated LLM APIs for contextual explanation generation, showing understanding of how generative AI enhances search results